Land 11: Lymphveil Moors

Vessels, lymph, and immunity · OpenStax Ch. 20 and 21

Key ideas

Vessel walls and vessel types

- Three layers: tunica intima (endothelium), tunica media (smooth muscle, elastic fibers), tunica externa (connective tissue).
- Elastic arteries near the heart stretch and recoil; muscular arteries distribute blood to organs.
- Arterioles are the main resistance vessels and control flow into capillary beds.
- Capillaries: continuous (most tissues, brain), fenestrated (kidney, intestine, endocrine glands), sinusoid (liver, spleen, marrow).
- Veins have thin walls, large lumens and valves; systemic veins and venules hold about 64 percent of blood volume.

Venous return and blood pressure

- Venous return relies on valves, the skeletal muscle pump and the respiratory pump.
- Pulse pressure equals systolic minus diastolic; 120/80 gives 40 mm Hg.
- MAP equals diastolic plus one third of pulse pressure; 120/80 gives about 93 mm Hg.
- Flow rises with the pressure gradient and falls as resistance rises.
- Resistance increases with viscosity and vessel length and varies with 1 divided by radius to the fourth power, so halving radius raises resistance 16 times.

Capillary exchange and edema

- Capillary hydrostatic pressure pushes fluid out; blood colloid osmotic pressure (albumin) pulls fluid in, about 25 mm Hg.
- Net filtration pressure equals hydrostatic pressure minus osmotic pressure; arterial end about 35 minus 25 gives 10 mm Hg out.
- At the venous end hydrostatic pressure is about 18 mm Hg, so net reabsorption is about 7 mm Hg.
- Lymphatic vessels return the small daily excess of filtered fluid.
- Edema results from high capillary pressure (heart failure), low plasma protein (liver failure), leaky capillaries (inflammation) or blocked lymph.

Regulating blood pressure

- The cardiovascular center in the medulla adjusts heart rate, contractility and vessel diameter.
- Baroreceptors in the carotid sinuses and aortic arch sense stretch; low pressure triggers sympathetic output.
- Chemoreceptors sense high CO2, low pH and low O2 and raise cardiac output and vasoconstriction.
- Renin converts angiotensinogen to angiotensin I; ACE in the lungs makes angiotensin II, which constricts vessels and stimulates aldosterone release.
- ADH retains water, epinephrine raises heart rate and contractility, and ANP lowers volume and pressure.

Shock, vessels and fetal circulation

- Shock types: hypovolemic (fluid loss), cardiogenic (pump failure), vascular or distributive (septic, anaphylactic, neurogenic) and obstructive.
- Hypertension is chronically elevated arterial pressure (exact thresholds vary by guideline); over time it damages vessels, heart and kidneys.
- Aortic arch branches: brachiocephalic trunk, left common carotid, left subclavian.
- The hepatic portal vein carries blood from the digestive organs and spleen to the liver for processing.
- Fetal shunts: foramen ovale (right to left atrium), ductus arteriosus (pulmonary trunk to aorta), ductus venosus (bypasses liver).

Lymphatic system

- Lymph moves by skeletal muscle and respiratory pumps, smooth muscle in vessel walls, and valves.
- The thoracic duct drains most of the body into the left subclavian vein; the right lymphatic duct drains the upper right quadrant.
- Primary lymphoid organs: red bone marrow (B cells mature) and thymus (T cells mature).
- Secondary organs: lymph nodes filter lymph, the spleen filters blood and removes old red cells, tonsils guard the pharynx.
- MALT, such as Peyer's patches in the ileum, protects mucosal surfaces.

Innate immunity

- First line: skin, mucous membranes and chemical secretions.
- Phagocytes: neutrophils arrive first; macrophages clean up and present antigen.
- Natural killer cells destroy virus infected and tumor cells without prior sensitization.
- Inflammation signs: redness, heat, swelling, pain; histamine increases capillary permeability.
- Fever raises the hypothalamic set point; interferons block viral spread; complement causes opsonization, inflammation and lysis by the membrane attack complex.

Adaptive immunity and antibodies

- MHC I on all nucleated cells is read by CD8 cytotoxic T cells; MHC II on antigen presenting cells is read by CD4 helper T cells.
- Helper T cells activate B cells and cytotoxic T cells; regulatory T cells suppress responses.
- Activated B cells become plasma cells that secrete antibody and memory cells for later exposure.
- IgG: most abundant, crosses placenta. IgM: pentamer, first made. IgA: secretions and milk. IgE: allergy and parasites. IgD: B cell receptor.
- Antibodies neutralize, agglutinate, opsonize and activate complement; the secondary response is faster, larger and mostly IgG.

Immunity types and disorders

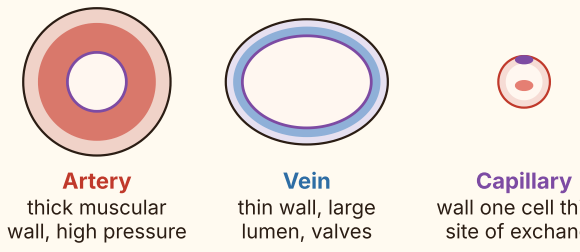
- Active immunity makes memory; passive immunity uses borrowed antibodies and is short lived.
- Natural active: infection. Artificial active: vaccine. Natural passive: placenta or milk. Artificial passive: immune globulin injection.
- Hypersensitivity: type I IgE and mast cells, type II antibodies on cells, type III immune complexes, type IV delayed T cells.
- Autoimmunity is loss of self tolerance, as in type 1 diabetes, rheumatoid arthritis and lupus.
- HIV destroys CD4 helper T cells; transplant rejection occurs when recipient T cells attack donor MHC.

Villager wisdom

- Arterioles are the sluice gates of the moor. Narrow one even a little and resistance climbs sharply, because radius matters most of all.
- In a capillary, blood pressure pushes fluid out near the arterial end, and albumin pulls most of it back near the venous end. Lymph vessels carry home the rest.
- When pressure drops, the baroreceptors in your carotid sinus and aortic arch sound the alarm, and the heart speeds up while vessels tighten.
- Innate defenders like neutrophils, natural killer cells and complement strike fast at anything foreign. B and T cells are slower but remember every foe.
- A vaccine shows your memory cells a foe's face without the illness, so the next meeting brings a swift, strong secondary response.

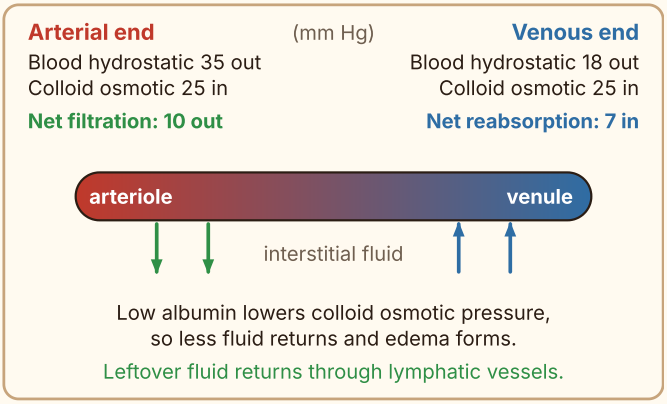
Diagrams

Arteries, veins and capillaries



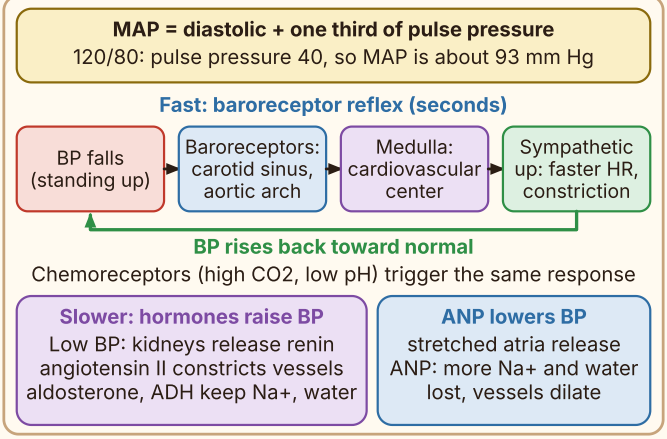
Arteries have thick muscular walls, veins have thin walls, large lumens and valves, and capillaries are one cell thick; arterioles set most of the resistance.

Capillary filtration and reabsorption



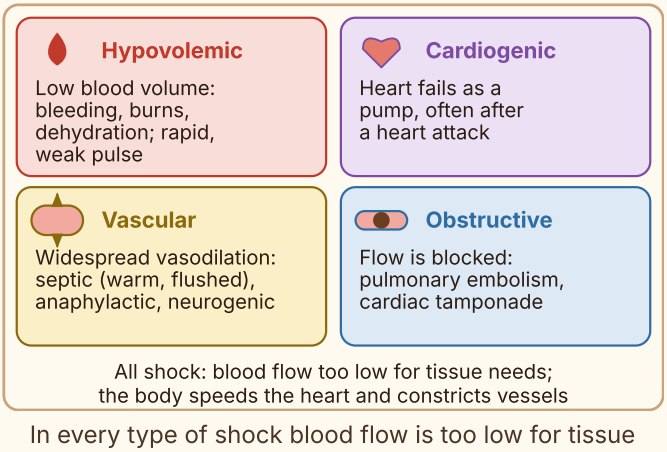
At the arterial end blood hydrostatic pressure (35 mm Hg) beats colloid osmotic pressure (25) so fluid filters out; at the venous end (18 versus 25) fluid returns.

Regulation of blood pressure



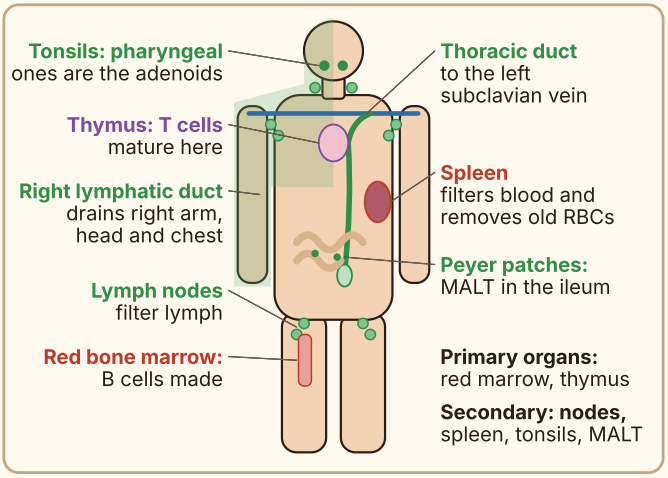
MAP equals diastolic pressure plus one third of the pulse pressure; the baroreceptor reflex corrects sudden drops in seconds, while hormones such as angiotensin II, aldosterone, ADH and ANP adjust pressure over hours.

Types of shock



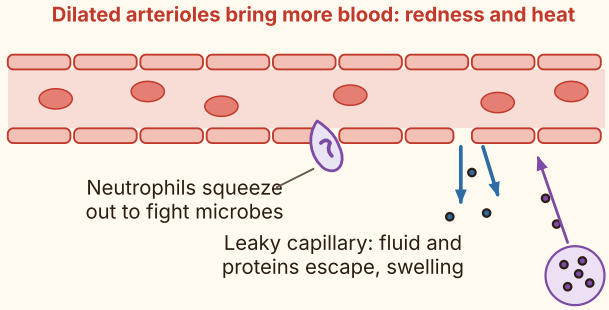
In every type of shock blood flow is too low for tissue needs; hypovolemic shock follows blood or fluid loss, and septic shock is vascular shock from widespread vasodilation.

Lymphatic system and lymphoid organs



T cells mature in the thymus and B cells in red marrow; lymph nodes filter lymph and the spleen filters blood. The right lymphatic duct drains the right arm, head and chest; the thoracic duct drains everything else into the left subclavian vein.

Inflammation



Histamine dilates arterioles (redness, heat) and makes capillaries leaky, so fluid and proteins escape (swelling) and neutrophils move into the tissue.

Innate and adaptive immunity

Innate
fast, nonspecific, no memory

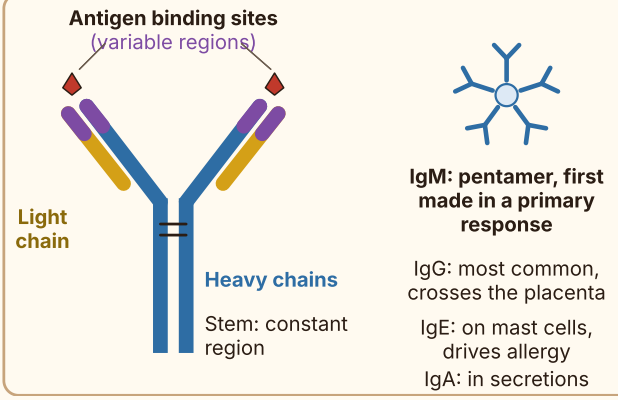
- Skin and mucous barriers
- Neutrophils, macrophages
- NK cells kill infected and tumor cells
- Complement lyses bacteria
- Inflammation and fever

Adaptive
specific, has memory

- B cells become plasma cells that make antibodies
- Helper T: CD4, sees antigen on MHC II
- Cytotoxic T: CD8, sees antigen on MHC I
- Memory B and T cells give a fast second response

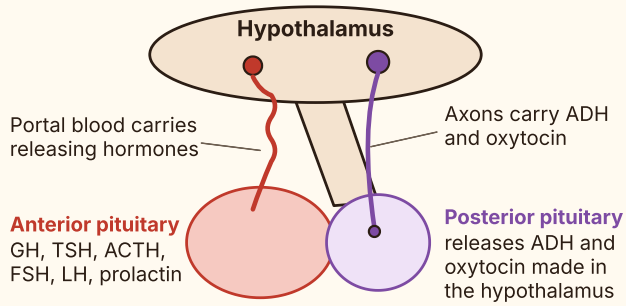
Innate defenses act fast against any invader; adaptive immunity is specific, uses B and T lymphocytes, and keeps memory cells for a faster second response.

Antibody structure



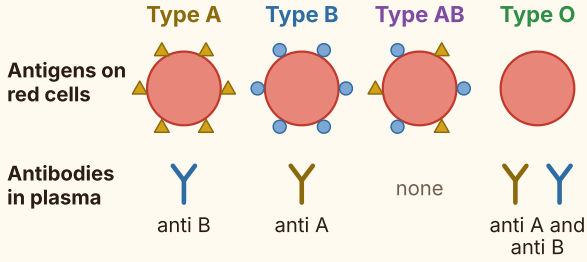
An antibody has two heavy and two light chains; the variable tips bind one specific antigen, and the constant stem sets its class (IgM, IgG, IgE, IgA).

Hypothalamus and pituitary



Hypothalamic releasing hormones reach the anterior pituitary through portal blood; ADH and oxytocin are made in the hypothalamus and released from the posterior pituitary.

ABO and Rh blood groups

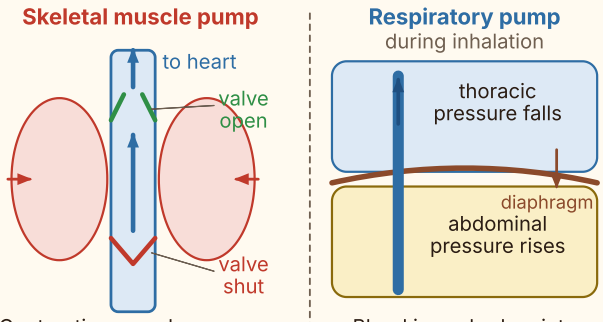


Rh: an Rh negative mother exposed to Rh positive blood makes anti D. Anti D IgG can cross the placenta and attack red cells of a later Rh positive fetus.

RhoGAM given to the mother prevents this.

Plasma holds antibodies against the ABO antigens a person lacks, so type A blood carries anti B and attacks type B donor cells.

Venous return

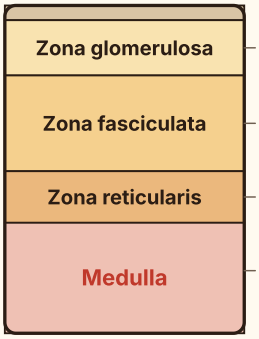


Contracting muscles squeeze veins; valves stop backflow

Blood is pushed up into the thoracic veins

Contracting skeletal muscles squeeze deep veins while valves block backflow, and during inhalation falling thoracic pressure and rising abdominal pressure push blood toward the heart.

Adrenal gland zones



Mineralocorticoids: aldosterone (salt)

Glucocorticoids: cortisol (sugar)

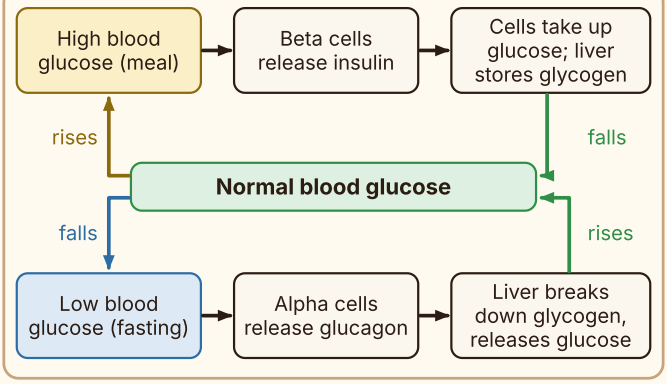
Androgens (sex hormones)

Epinephrine and norepinephrine; cells are modified sympathetic neurons

Too much cortisol: Cushing syndrome

The adrenal cortex makes aldosterone (glomerulosa), cortisol (fasciculata) and androgens (reticularis); the medulla, modified sympathetic neurons, releases epinephrine and norepinephrine.

Insulin and glucagon



Insulin from beta cells lowers blood glucose by moving it into cells and glycogen; glucagon from alpha cells raises it by breaking down liver glycogen.

Self check

1. **An infant receives antibodies through breast milk. Which type of immunity is this?**
    A. Natural active immunity
    B. Natural passive immunity
    C. Artificial passive immunity
    D. Artificial active immunity
2. **A cytotoxic T cell recognizes a virus infected body cell by detecting viral peptides displayed on:**
    A. MHC class II
    B. MHC class I
    C. IgM receptors
    D. Complement C3b
3. **Which leukocyte is the first recruited from the blood in large numbers to a site of acute bacterial infection?**
    A. Monocyte
    B. Neutrophil
    C. Eosinophil
    D. Basophil
4. **Lymph nodes protect the body mainly by:**
    A. Removing old red cells from the blood
    B. Producing lymph from blood plasma
    C. Filtering lymph so antigens meet lymphocytes
    D. Serving as the site of T cell maturation
5. **At the arterial end of a capillary, fluid filters out into the tissues mainly because:**
    A. Osmotic pressure exceeds hydrostatic pressure
    B. Plasma proteins actively pump fluid out
    C. Lymph vessels push fluid outward
    D. Hydrostatic pressure exceeds osmotic pressure
6. **Which antibody class appears mostly as a receptor on naive B cell surfaces, with very little found in plasma?**
    A. IgD
    B. IgM
    C. IgG
    D. IgA
7. **Which hormone released by the posterior pituitary raises blood volume by increasing water reabsorption in the kidneys?**
    A. Atrial natriuretic peptide
    B. Aldosterone
    C. Antidiuretic hormone (ADH)
    D. Renin

8. **The brachiocephalic trunk divides into which two arteries?**
    A. Left common carotid and left subclavian
    B. Right common carotid and right subclavian
    C. Right and left vertebral arteries
    D. Right axillary and right brachial
9. **After major blood loss, a patient has low blood pressure and a rapid, weak pulse. Which type of shock is this?**
    A. Hypovolemic shock
    B. Cardiogenic shock
    C. Anaphylactic shock
    D. Neurogenic shock
10. **A patient with liver failure has low plasma albumin and swollen ankles. Why does edema develop?**
    A. Capillary hydrostatic pressure falls
    B. Colloid osmotic pressure falls, so less fluid returns
    C. Lymph flow becomes excessive
    D. Arterioles constrict and trap fluid
11. **Which vessel carries nutrient rich blood from the digestive organs to the liver?**
    A. Inferior vena cava
    B. Hepatic portal vein
    C. Hepatic artery proper
    D. Renal vein
12. **When a person stands up quickly and blood pressure drops, the baroreceptor reflex responds by:**
    A. Slowing heart rate and dilating arterioles
    B. Increasing parasympathetic output to the heart
    C. Raising heart rate and constricting arterioles
    D. Releasing atrial natriuretic peptide

Answer key

1. **B.** Natural passive immunity. The antibodies were made by the mother and acquired through a normal life event, so the immunity is natural and passive. It is not active because the infant's own B cells did not make them.
2. **B.** MHC class I. Nearly all nucleated cells display internal peptides on MHC class I, which CD8 cytotoxic T cells inspect. MHC class II is found only on antigen presenting cells and is read by CD4 helper T cells.
3. **B.** Neutrophil. Neutrophils arrive within hours and phagocytize bacteria, making up most of the early response. Monocytes migrate later and become macrophages that clean up debris.
4. **C.** Filtering lymph so antigens meet lymphocytes. As lymph percolates through a node, macrophages trap debris and lymphocytes encounter antigen. Removing aged red cells is a function of the spleen, which filters blood.
5. **D.** Hydrostatic pressure exceeds osmotic pressure. Near the arterial end, capillary hydrostatic pressure (about 35 mm Hg) is greater than blood colloid osmotic pressure (about 25 mm Hg), so the net force pushes fluid out. Near the venous end the hydrostatic pressure has fallen below the osmotic pull, so the balance favors reabsorption instead.
6. **A.** IgD. IgD serves as a B cell antigen receptor and is barely detectable in serum. IgM is also on naive B cells, but it is plentiful in plasma as the pentamer secreted first in a primary response.

7. **C.** Antidiuretic hormone (ADH). ADH, also called vasopressin, increases water reabsorption and in high amounts constricts vessels. Aldosterone also raises blood volume, but it comes from the adrenal cortex and acts by retaining sodium.
8. **B.** Right common carotid and right subclavian. The brachiocephalic trunk exists only on the right side. The left common carotid and left subclavian arteries branch directly from the aortic arch.
9. **A.** Hypovolemic shock. Hemorrhage lowers blood volume, which reduces venous return and cardiac output, and the heart races to compensate. Cardiogenic shock also causes low pressure, but it comes from a failing pump such as after a heart attack, not from lost volume.
10. **B.** Colloid osmotic pressure falls, so less fluid returns. Albumin from the liver creates most of the osmotic pull that draws fluid back into capillaries, so low albumin means more fluid stays in the tissues. A fall in hydrostatic pressure would actually reduce filtration and work against edema.
11. **B.** Hepatic portal vein. The hepatic portal vein delivers blood from the stomach, intestines, spleen and pancreas to the liver so it can process absorbed nutrients and toxins. The hepatic artery proper also supplies the liver, but it brings oxygenated blood from the aorta, not blood from the gut.
12. **C.** Raising heart rate and constricting arterioles. Reduced stretch in the carotid sinus and aortic arch lowers baroreceptor firing, so the medulla increases sympathetic output, which speeds the heart and constricts vessels to restore pressure. Increasing parasympathetic output would slow the heart and make the drop worse.